

Novel Double Rate Driving Pixel Structure for High-Performance 4K 165Hz OLED TVs with Reduced Number of Source Driver ICs

Won-Ho Lee, Young-Min Jeong, Sang-Hoon Jung, Jun-Young Yang, Soo-Young Yoon

Advanced Technology Laboratory, LG. Display, LG Science Park, Seoul, Korea

Abstract

We have developed innovative double rate driving (DRD) pixel structure designs for large-area, high-refresh-rate OLED TVs. The novel DRD pixel structure adopts a link branch to connect separated data lines, ensuring that each data line is connected to pixels of the same color. The signal in the data line of this structure is similar to that of a single rate driving (SRD) pixel structure, which offers advantages in charging efficiency compared to conventional DRD pixel structures, making it favorable for high-refresh-rate products. Additionally, the source driver IC (SD-IC) temperature was reduced by approximately 50% compared to the conventional DRD pixel structure.

Keywords

double rate driving pixel structure, high-refresh-rate, source driver IC temperature, OLEDs.

1. Introduction

The demand for advancements in TV specifications, such as larger screen sizes, higher brightness, and higher refresh rates, has been consistently increasing. In response, as shown in Figure 1, the display device industry has researched DRD pixel structures that can halve the number of SD-ICs to enhance price competitiveness. For large products like 4K resolution TVs, DRD pixel structure technology capable of high-speed operation without image quality degradation is essential.

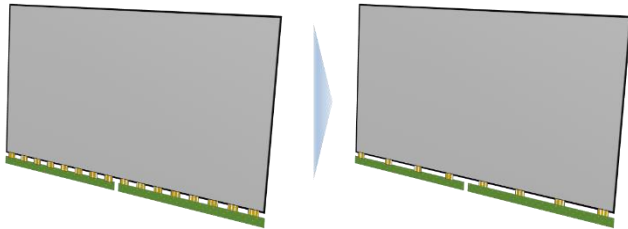


Figure 1. Reduction of the number of SD-IC via DRD pixel structure technology

However, previous attempts to develop such technology faced challenges, including performance constraints such as image quality degradation and high SD-IC temperatures, which hindered the successful expansion of such products. This paper introduces the concept of a novel DRD pixel structure that overcomes the constraints, along with its enhanced operating principle and mechanism.

2. Results and discussion

2.1 Novel dual gate pixel structure design

Unlike LCDs, OLEDs do not have pixel polarity, enabling DC driving without data inversion in solid patterns of monochromatic or mixed colors. However, as shown in Figure 2, in OLEDs with a

conventional DRD pixel structure that connects adjacent different color pixels and data lines, data transitions still occur, leading to reduced charging efficiency and image quality degradation. Reduced charging efficiency leads to lower brightness and poorer brightness uniformity.

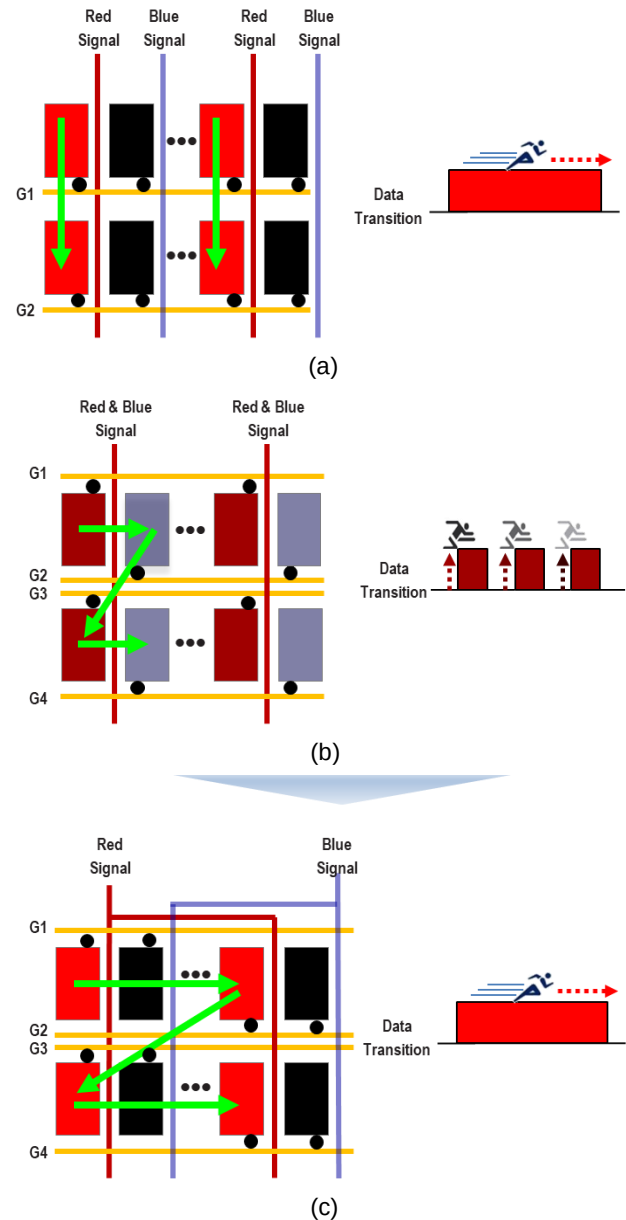


Figure 2. Schematic of pixel structures: (a) Conventional SRD pixel structure (b) Conventional DRD pixel structure (c) Novel DRD pixel structure

Moreover, as shown in Figure 3, in the conventional DRD pixel structure, the data signal can be mixed with other colors in monochrome patterns, causing color mixing and potentially degrading color characteristics. The high number of data transitions in the conventional DRD pixel structure also leads to unfavorable conditions for SD-IC temperature.

To resolve these issues, we branched two data lines of the same color in the link section, enabling DC driving without data switching in monochrome patterns, similar to the SRD pixel structure. DC driving of identical color pixels in solid patterns improved charging characteristics. Additionally, the elimination of data transitions reduced SD-IC temperature increases.

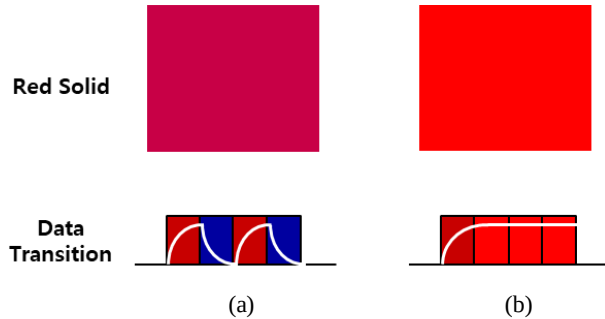


Figure 3. Red solid pattern comparison for conventional and novel DRD pixel structures: (a) Conventional DRD pixel structure (b) Novel DRD pixel structure

2.2 Improving data transition counts in special patterns

Compared to SRD pixel structures, DRD pixel structures halve the charge time per pixel, thereby doubling the number of data transitions in special patterns like dot patterns. However, in the new structure, pixels of the same color are connected via a link branch. As shown in Figures 4 and 5, this design maintains the same number of data transitions as the SRD pixel structure, thereby achieving a charging efficiency of over 80% across the entire grayscale range in dot pattern. Consequently, the SD-IC temperature was reduced by approximately 50% compared to the adjacent different color pixel-sharing structure. Figure 6 shows the SD-IC temperature for SRD pixel structure and novel DRD pixel structure.

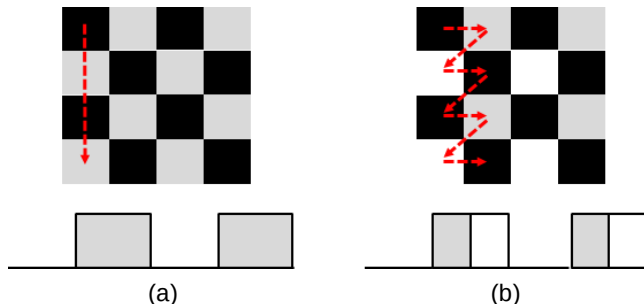


Figure 4. Comparison of data transition for SRD pixel structure and novel DRD pixel structure in dot pattern: (a) SRD pixel structure data driving (b) Novel DRD pixel structure data driving

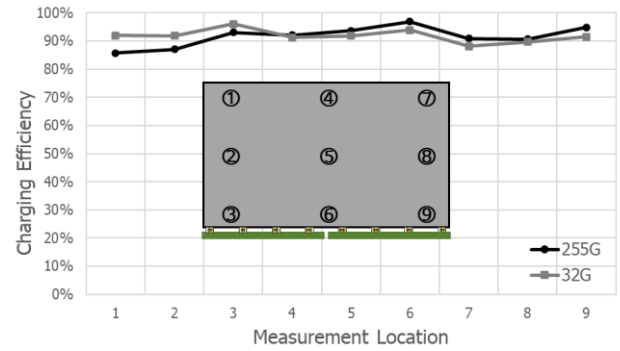


Figure 5. Charging efficiency by panel measurement location for novel DRD pixel structure

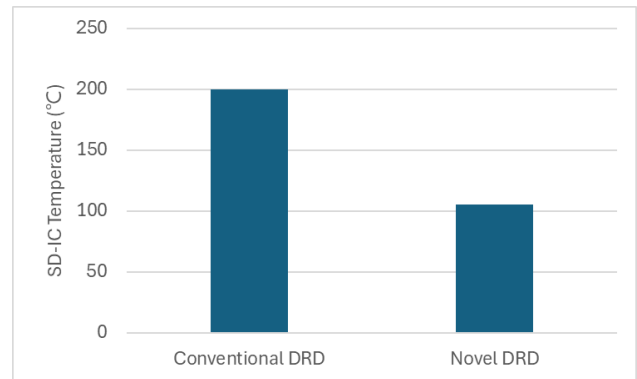


Figure 6. Comparison of SD-IC temperature for SRD pixel structure and novel DRD pixel structure

2.3 Alternate frame driving

When transitioning from black to white such as h1by1 pattern or dot pattern, luminance differences arise due to variations in charging efficiency between the first and second pixels. To resolve this issue, we improved uniformity by alternately driving the Z-Driving and S-Driving between frames, as illustrated in Figure 7.

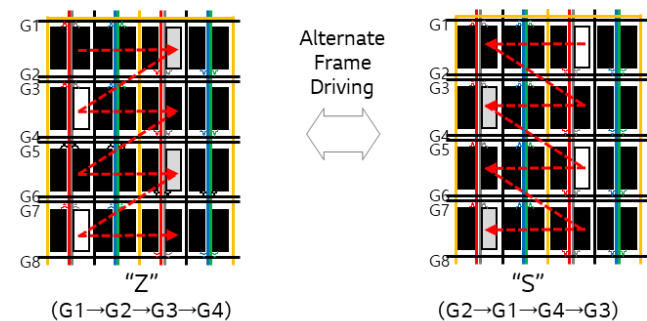


Figure 7. Alternate frame driving (AFD)

Here, Z-Driving refers to the gate signal activation sequence $G1 \rightarrow G2 \rightarrow G3 \rightarrow G4$, which results in a Z-shaped pixel drive sequence. Correspondingly, S-Driving operates in the sequence $G2 \rightarrow G1 \rightarrow G4 \rightarrow G3$, yielding an S-shaped pixel drive sequence. Z-Driving and S-Driving alternate between frames, averaging the

weak charged pixels and strong charged pixels per frame, thereby improving image quality characteristics and luminance uniformity.

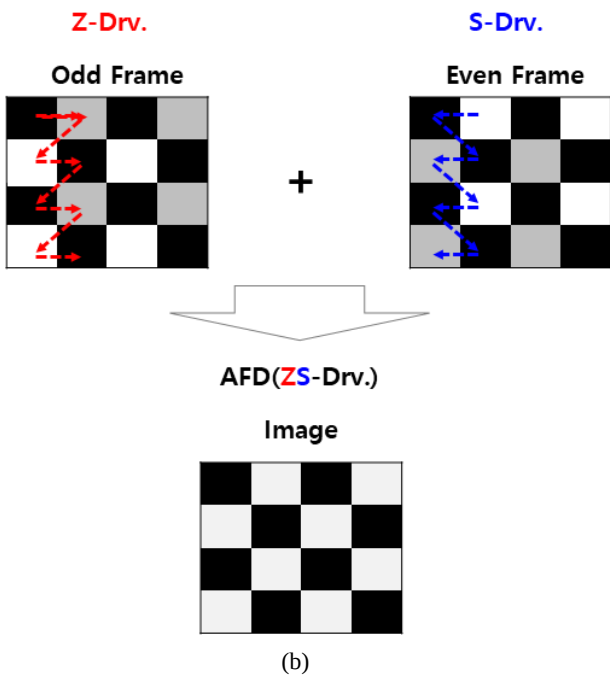
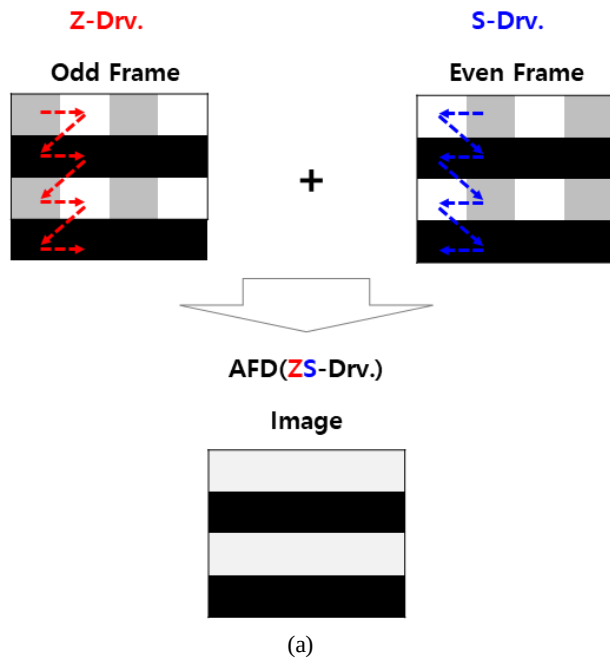


Figure 8. AFD for image quality improvement in special patterns such as (a) h1b1 pattern or (b) dot pattern

3. Summary

This paper presents the development of an innovative double rate driving (DRD) pixel structure designed for large-area, high-refresh-rate OLED TVs. The proposed DRD pixel structure introduces a link branch that connects separated data lines, ensuring that each data line is dedicated to pixels of the same color. This design mimics the signal behavior of a single rate driving (SRD)

pixel structure, thereby improving charging efficiency and reducing image quality degradation, which are common issues in conventional DRD structures.

1) Enhanced Charging Efficiency: By enabling DC driving without data switching in solid patterns of monochromatic or mixed colors, the novel DRD structure achieves charging efficiency comparable to SRD structures. This is particularly beneficial for maintaining brightness and uniformity in solid color patterns.

2) Improved Performance in Special Patterns: The structure maintains the same number of data transitions as SRD in special patterns like dot patterns, achieving over 80% charging efficiency across the entire grayscale range.

3) Reduced Source Driver IC (SD-IC) Temperature: The new design significantly reduces the number of data transitions, leading to about 50% decrease in SD-IC temperature compared to conventional DRD structures. This improvement is crucial for the reliability and longevity of the display system.

4) Alternate Frame Driving (AFD): To address luminance differences in patterns such as h1b1 or dot patterns, the paper introduces an alternate frame driving technique. This method alternates between Z-Driving and S-Driving sequences, improving image quality and luminance uniformity by averaging the charging characteristics of pixels.

The proposed DRD pixel structure not only enhances the performance of high-refresh-rate OLED TVs but also enables a reduction in the number of SD-ICs by half, significantly improving cost competitiveness for 42"-77" 4K 165Hz TVs. This innovation addresses previous limitations in DRD technology, offering a promising solution for the next generation of large-area OLED displays.

4. Conclusion

The application of the novel DRD pixel structure significantly improved image quality and SD-IC temperature performance compared to conventional DRD pixel structures. This breakthrough enabled the reduction of SD-ICs by half for the first time globally, enhancing product competitiveness for 42"-77" 4K 165Hz TVs.

5. References

- [1] Liqing Liao, et al., "Novel dual gate pixel design with high image quality and low power consumption for LCDs", SID 2023, pp. 1395~1398.
- [2] Sai Chang Yun, et al., "43.2: New Driving Method for Low Logic Power Consumption in TFT-LCDs". SID Symposium Digest of Technical Papers, 2010, 41: pp. 619-621.
- [3] Dongchuan Chen et al., "29-4: Research on Dual-Gate Pixel Architecture Enabling High-PPI and High-Refresh-Rate LCD Panels". SID Symposium Digest of Technical Papers, 2025, 56: pp. 267-270.